

Contents

1 Chapter Two: "Text Analysis"	1
1.1 Doubt	1
1.2 Falsifiable	2
1.3 Iteration in gender	12
1.4 Iteration in python	18
1.5 TODO Fluidity	26
1.6 Conclusion	33

1 Chapter Two: "Text Analysis"

1.1 Doubt

1 The novel *Orlando: A Biography* (1928), by Virginia Woolf, famously opens with an assertive gender designation followed by an immediate qualification: "He—for there could be no doubt of his sex, though the fashion of the time did something to disguise it" (11).

2 In order to perform quantitative text analysis on this text, the standard tasks of "pre-processing" would evacuate this sentence of its gender qualification. Text analysis, a process which involves calculating and visualizing textual patterns, requires transforming the source text into a computable format. Pre-processing the text (also called "cleaning" or "regularizing" the text) would strip capitalized words, punctuation, "stop words" (such as articles and prepositions), and inflections in word endings, all of which are deemed to be semantically minor, and wouldn't affect the analysis of more substantial features like nouns, verbs, adverbs, and adjectives.

3 Cleaning the first sentence not only would strip it of its pronouns, including the gender assertion in the first word, "He." It would also cut the em dash immediately following this "He," signaling the entrance of the narrator and his conspicuous certitude: "—for there could be no doubt of his sex..." Only the following list of words would remain:

‘could’, ‘doubt’, ‘sex’, ‘though’, ‘fashion’, ‘time’, ‘something’,
‘disguise’

4 Text analysis requires removing such "minor" elements of language in order to find connections among more "substantial" terms. None of these features, however, are minor in a text about a fictional biography of a 16th-century English nobleman who undergoes a sex change. Orlando's gender,

as many critics point out, develops along with his story.¹ His trials and tribulations experienced as a man are resolved when he becomes a woman, and she experiences a form of self-actualization that not only transcends time but also, for a moment, stops it. If Orlando has a thesis about gender, it is this—gender is what one makes of it. It relies fantasy, imagination, and most importantly, language, in order to come into being.

5 This chapter examines how quantitative text analysis works to analyze gender by using Woolf's *Orlando* as a test case. It explores an experimental approach to text analysis that deconstructs gender binaries by drawing connections between computer programming and gender theory—Judith Butler's theory of gender performativity. This analysis surfaces a mechanism common to both text analysis and gender performativity, which is *iteration*. There is an iterative method that works on both text analysis and on gender which, I argue, can be harnessed to resignify text analysis procedures. This chapter proposes a text analysis procedure that iterates through various gender terms in *Orlando*, to re-signify their meaning within binary constraints. The goal here, as with Butler's own work, is to work within constrictive structures, like the gender binary, in order to unmake them from the inside.

1.2 Falsifiable

1

Text analysis has always been driven by great ambitions. Matthew Jockers's book, *Macroanalysis: Digital Methods and Literary History* (2013), "ask[s] very big questions. . . regarding how literary trends are employed over time, across periods, within regions"; Ted Underwood's book, *Distant Horizons: Digital Evidence and Literary Change* (2020) traces the "larger arcs of literary change that have remained hidden from us by their sheer scale." Most recently, with the advent of text generation methods, Leif Weatherby's book, *Language Machines: Cultural AI and the End of Remainder Humanism* explains that language models which "'pretrain' on massive corpuses of text data—trillions of words scraped mostly from the internet", bring language into "the age of its algorithmic reproducibility".

Predating (and perhaps inspiring) much of this work in text analysis is an even greater ambition by from a literary scholar named Franco Moretti. Morretti, who introduced many both within and without Literary Studies to text analysis, published a polemical essays on "distant reading" in a collection of the same name in 2008. In 2017, his research caught the attention

¹Criticism of the novel about gender and story.

of the *New York Times*, who published a profile detailing his "radical cry" for a new kind of literary analysis. Distant reading, according to Moretti, involves counting up textual features to derive textual patterns and make inferences. It is a kind of analysis where "distance" as Moretti explains, "is a condition of knowledge" (48).

2

In the wake of his 2017 profile by *New York Times*, three students accused Moretti of sexual crimes. The accusations ranged from unsolicited touching to rape. One of his accusers was a first-year graduate student in English.

When I was a graduate student at UC Berkeley in 1984-85, my then-professor Franco Moretti sexually stalked, pressured, and raped me. He specifically said to me, 'You American girls say no when you mean yes.'[...] I reported him to the Title IX officer, who was then Frances Ferguson, Ph.D. She was a friend of his and urged me not to make a report[...] I told Franco about my conversation with Ferguson, and he threatened to ruin my career if I pressed charges against him. He said he had powerful friends who were attorneys who would ruin my name. I remained silent for all these years because I was in academia. I have told a number of people about it privately, however. These are up-standing, well-known professors of history and English at other institutions, who would certainly corroborate my story. (Kimberly Latta "Op-Ed: What Happened?" 2017).

When the women came forward, Stanford went no further than claiming they would review the cases. There has been no record of an inquiry, investigation, or any repercussions for Moretti (Liu, Knowles, "Harassment, assault allegations against Moretti span three campuses"). He remains an Emeritus Professor at the same institution.

3

Moretti's writing has an easy, conversational flow. He uses short sentences, but with varied syntactic structures and frequent commas, giving his prose a sense of rhythm. He also weaves in references across fields of knowledge as disparate as evolutionary theory, systems theory, and Marxism. Despite this intellectual baggage, his arguments move at a light clip, and seem to circle methodically around a drain so that, as long as you're following him, there appears to be no alternative. The effect is entrancing.

But there is something suspect about his confidence. Not just in his ambition, which is shared by several others, but the way he offers his argu-

ments. He transforms inferences, premises, and hypotheses into declarative statements. Paragraphs begin with sentences like,

So, the market selects the canon. But how? (69).

Yes, forms *can* move in several directions. But *do* they? (emphasis original, 112).

So. As novelistic forms travel through the literary system, their plots are (largely) preserved, while their styles are (partly) lost—and replaced by 'local' ones (134).

What did the birth of a consumer society mean for the European novel? More novels, less attention (175).

The major metamorphosis of eighteenth-century titles is simple: in the space of two generations, they become much, much shorter (182).

As the market expands, titles contract; as they do that, they learn to compress meaning; and as they do *that*, they develop special 'signals' to place books in the right market niche (204).

These quotes, taken from mostly different essays in *Distant Reading*, show Moretti's characteristic way of stating a claim or hypothesis as a conclusion, as if the point is already self-evident. To be fair, many of these points do synthesize arguments and evidence from previous paragraphs. They are, some of them, more summary of previous points than completely new ones. But it is the way they are presented here that strikes me. Their apparent solidity, the lack of a subjective "I" to ground them, the presumptuousness and ease with which they make room for themselves in the world. It is not confidence, it is something more than confidence, which felt exciting at the same time that it felt dangerously familiar.

4

One of Moretti's earliest essays, "The Soul and the Harpy" sows the seeds that would eventually grow into distant reading. Published in 1983, two years before the date of the first report of assault of against him, this essay lays out new methodological standards for literary criticism. In it, he appears to be very concerned about the apparently unstable methodology of close reading, which drives contemporary literary criticism; he betrays a deep suspicion about this and other analytical practices that "multiply, rather than reduce, the obstacles every social science encounters when it tries to give itself a testable foundation" ("Soul" 22). He likens the literary

critic to "Narcissus", "whose only pleasure lay[s] in contemplating his own reflection ("Soul" 14).

Moretti proposes an alternative methodology, a "falsifiable criticism." This method grounds the critical process in hypotheses to "test" literary interpretations, pursuing answers that are "coherent, univocal, and complete" ("Soul" 21). But a close look at his reasoning suggests that the goal may not be so much to verify interpretations, but to falsify them.

"The day criticism gives up the battle cry 'it is possible to interpret this element in the following way,' to replace it with the much more prosaic 'the following interpretation is impossible for such and such a reason,' it will have taken a huge step forward on the road of methodological solidity" ("Soul" 22).

To squash the possibility of alternative interpretations. For literary criticism to achieve a means of attaining irrefutable conclusions. Moretti here implicitly aligns literary criticism with methodologies from the sciences, known as reproducible methods.² In those disciplines, reproducibility means that the analytical procedure will yield the same result.³

5

Moretti's nascent idea of falsifiable criticism finds a methodology in quantification, that is, methodologies based on the counting of textual features. Quantitification, according to Moretti, simplifies text into workable units. Language in all its complexity is flattened into features which can be counted and then graphed. As he explains, "fewer elements, hence a sharper sense of their overall interconnection" (*Graphs, Maps, Trees* 1). These fewer elements, he argues, enable ambitious interpretations: "If we want to understand the system in its entirety, we must accept losing something. We always pay a price for theoretical knowledge" (*Distant Reading* 49). What is lost, textual detail and specificity, is a worthwhile price for the potential reward. "The more ambitious the project," Moretti explains, "the greater must the distance be" (*Distant Reading* 48).

6

²Sarah Cohen-Boulakia, Khalid Belhajjame, Olivier Collin, Jérôme Chopard, Christine Froidevaux, Alban Gaignard, Konrad Hinsén, Pierre Larmande, Yvan Le Bras, Frédéric Lemoine, Fabien Mareuil, Hervé Ménager, Christophe Pradal, Christophe Blanchet, Scientific workflows for computational reproducibility in the life sciences: Status, challenges and opportunities, *Future Generation Computer Systems*, Volume 75, 2017, Pages 284-298, ISSN 0167-739X <https://doi.org/10.1016/j.future.2017.01.012>.

³Reproducibility and transparency: what's going on and how can we help. *Nat Commun.* 2025 Jan 27;16(1):1082. doi: 10.1038/s41467-024-54614-2. PMID: 39870612; PMCID: PMC11772749.

In one famous essay from *Distant Reading*, "Style, Inc.," Moretti reduces entire novels to just their titles. Here, he explores what titles, which get shorter and shorter over the 18th century, might reveal about trends in the book market. Moretti describes his analytical process:

[F]irst, I describe a major metamorphosis of eighteenth-century titles, and try to explain its causes; next, I suggest how a new type of title that emerged around 1800 may have changed what readers expected of novels; and finally, I make a little attempt at quantitative stylistics, examining some strategies by which titles point to specific genres. Three sections, three pieces in the large puzzle of the literary field. (*Distant Reading* 181-2; emphasis mine)

Moretti's critical method involves posing hypotheses, assembling and analyzing data, making inferences, and occasionally, reframing the original hypotheses. Yet, he repeatedly understates his interpretive moves at each stage of his analysis: "try to explain," "suggest," "make a little attempt." The diminutive language here not only belies the presumptions his arguments make, for example, that titles function as a proxy for market forces, "a code, in the market: half sign, half ad... where the novel as language meets the novel as community" (181). It also misrepresents the role that inferential thinking plays in the conclusions he draws:

As the market expands, titles contract; as they do that, they learn to compress meaning; and as they do *that*, they develop special 'signals' to place books in the right market niche. (204)

As Stephen Ramsay points out, Moretti presents his own insights as a description of reality, as if "data is presented to us... not as something that is also in need of interpretation" (Ramsay, *Reading Machines* 5). Moretti here performs an agential sleight-of-hand: diminishing his own interpretive role while amplifying the agency of inanimate phenomena like book titles. Titles may fluctuate, and as they do so, his analysis deftly shows, patterns emerge around the consolidation of nouns and adjectives. But they do not, on their own, *learn to compress meaning or develop special 'signals'*.

7

Distant reading deliberately overlooks details which, as some have aptly pointed out, are actually necessary to cultural critique. Details of text would express what has been traditionally minimized or marginalized. In an MLA panel titled "Distant Reading After Moretti," Lauren Klein explains,

[I]t's not a coincidence that distant reading does not deal well with gender, or with sexuality, or with race. Gender and sexuality and race are precisely the sorts of concepts that have been exposed and interrogated by attending to non-dominant subject positions. And like literary world systems, or "the great unread," the problems associated with these concepts, like sexism or racism, are also problems of scale, but they require an increased attention to, rather than a passing over, of the subject positions that are too easily (if at times unwittingly) occluded when taking a distant view.

Klein's own work tracing these elements, for example, in hidden networks within archival documents, proves how careful attention to what is minor or oblique can surface previously suppressed stories.⁴ In one project, Klein uses network analysis methods to bring forth the "ghost" of James Hemmings, an emancipated slave who figured in the correspondence of a famous slaveholder. By marking up oblique references in the letters of his former master (Thomas Jefferson), Klein "render[s] visible the archival silences implicit in our understanding of chattel slavery today" (665). Her work demonstrates that the details of Hemmings's life, which is deeply distributed in the correspondence data, requires a kind of reading that is close and careful.

8

In opposition to close and careful reading, the ambition to grasp large swaths of literary history may be a proxy for avoiding certain kinds of cultural critique. After all, Moretti says "we know how to read texts, now let's learn how *not* to read them" (*Distant Reading* 48). The proxy functions through an amplification and valorization of what is seemingly objective—for example, the counting of words. With word counts and their resulting graphs and charts doing the work of interpretation, one might finally *close down* close reading and the pesky critiques associated with it. One might finally approach a *falsifiable criticism*, to be able to say once and for all that "the following interpretation is impossible for such and such a reason" (Moretti, "Soul" 22).

9

This faith in the falsifiable—in the power of counting words—infects even those who disavow quantitative methods. In a detraction written by Nan Z. Da, which rallied the Computational Literary Studies debate when it

⁴Klein, Lauren F., "The Image of Absence: Archival Silence, Data Visualization, and James Hemmings," in *American Literature*, 85 (4): 661–688, 2013.

was released in 2020,⁵ Da vehemently opposes the use of quantification in critical interpretation. Da argues that quantitative methods, in fact, reveal a "fundamental mismatch between the statistical tools that are used and the objects to which they are applied"—that is, to literary objects (620, 601). She emphasizes her point by attempting to reproduce an experiment in "topic modeling," a machine learning method that generates a number of "topics," or keywords, from large collections of text. When she attempts to replicate a topic modeling experiment on her own computer, she produces different results.

10

Despite her oppositional stance toward quantitative tools, Da has something in common with those who champion them. She fundamentally *agrees* about the objectivity of their procedures. As Ben Schmidt writes about Da's stance, "Far more than anyone I've seen in any humanities article, she asserts that scientists do something arcane, powerful, and true" (Schmidt, "A computational critique of a computational critique of computational critique."). Despite their vastly different views on the role of quantitative methods for studying literature, Da and Moretti appear to agree that these methods ought to provide results that are, at the very least, reproducible.

- Katherine Bode's critique: "what seems a complete divergence of opinion obscures a fundamental agreement: that computation is separate from literary phenomena" ("What's wrong with Computational Literary Studies")

11

There are those who disagree about the apparent objectivity of reproducible methods. Ted Underwood, for example, explains that "algorithms are actually bad at being objective and rather good at absorbing human perspectives implicit in the evidence used to train them" ("Machine Learning and Human Perspective" 92). Underwood creates what he calls "perspectival models" of literary data, where models represent different "perspectives." In one project, on studying gender in novels, Underwood uses a binary classification algorithm to cluster terms that male and female writers tend to associate with either gender. From the results, he creates a chart which visualizes the differences between male and female writers and their treatment

⁵Expand on the computational literary studies field and debate.

At the time of its release, Da's article stoked many responses from literary scholars involved in text analysis. See Bode, "What's the Matter with Computational Literary Studies?"

of gender. In this chart (see Fig. 1), there are two diagonals. The diagonal moving from the bottom-left to top-right corner represents where the genders agree on gendered terms. The bottom-left shows agreement on terms used to describe the male gender ("pocket", "grinned"), and the top-right shows agreement on the female gender ("eyes," "mother"). The opposite diagonal, running from the top-left to bottom-right corners, shows disagreement. The top-left shows terms that each gender tends to use to describe their own gender ("wondered", "mind") and the bottom-right shows terms that each gender uses to describe the opposite gender ("meant", "kissed"). The chart, according to Underwood, shows two *perspectives* on gender—the male and the female. Rather than making claims to gender in an objective sense, it dramatizes a perspectival shift between the two genders.

12

Unfortunately, Underwood's methodology, as he himself admits, reinscribes the gender binary. This happens not only in the choice of model, a binary classification model which makes predictions on a yes/no or, in this case, a male/female scale (predicting whether a term will be more associated with a male or a female gender). It also happens with the choice in visualizing the results in a way that places male and female genders in opposition. The diagonal going from the bottom-left to the top-right, which finds similarity between the genders, reinforces a stereotypical active/passive binary (among other binaries) between men and women. Similarly, the diagonal going from the top-left to the bottom-right, although based on agreement between the genders and their relationships to themselves, nonetheless imposes an *opposition* between how male and female authors describe each other. The graph effectively instantiates its implicit assumption that male and female writers will oppose the other by visualizing this as a feature.

Not only does the diagram reify the gender binary, it effectively uses the gender binary *as a model* for proving the value of quantitative methods. This is an effective reversal of how data and models are supposed to work in support of each other. When explaining the diagram, Underwood says that "gender theorists will be frustrated by the binary structure of the diagram. . . I needed a simple picture, frankly, in order to explain how a quantitative model can be said to represent a perspective" ("Machine Learning and Human Perspective" 98). However, the "perspective" which he represents is one where the two genders are already diametrically opposed. The gender binary is a necessary presumption in order for the experiment to take place—a prior binary that then makes the choice of a binary-structured model appear appropriate. For his diagram to work, gender roles must have been already manufactured into opposition. In other words, binary gender is the model

through which his argument about perspectival modeling can make sense.

Add Bode's critique.

Bode's position is that QLS ought to begin by considering the types of gaps and biases in evidence. She wants critics to incorporate analyses from textual studies prior to running statistical tests. We should build QLS out of these areas of humanistic inquiry that center interpretation. There are certain processes in statistical analysis that maintain bias and unknown assumptions: sampling and comparison do not get an accurate picture if what we are searching for is unfindable or harder to find.

Responding to Ted Underwood's *Distant Horizons*. Critique on "perspectival modeling" as ignoring ethical issues that persist with scale.

- "Underwood's proposal to replace data modeling with sampling and comparison, alone or in concert with perspectival modeling, is ultimately inseparable from his view of the literary past as a single system of interlocking parts. But the literary past, and (quantitative) literary studies, are spaces of highly differentiated knowledges, opportunities, constraints, resources, and possibilities, and these must be considered in any arguments about, or propositions for the future of, the discipline" (116).

13

There are valid and even provocative reasons to study gender as a binary system. But the questions that drive such experiments must be differently derived. Questions like "how to distinguish one gender from another gender" effectively reify existing structures of the binary. Alternatively, other work uses text analysis to *deconstruct* social categories. In their introduction to *The Journal for Cultural Analytics*'s "Identity Issue," Susan Brown and Laura Mandell explain that, "The goal is to acknowledge the subjective effects of belonging to an identity constituted historically through oppression without believing that the identity itself exists independently from historical conditions" (6).

14

For example, Laura Mandell studies gender through genre. Mandell deploys the popular stylometry measurement, "Burrow's Delta," which visualizes the "distance" between writing styles by creating branches ("deltas") between different texts. She finds that Mary Wollstonecraft's style is shared across comparable male writers like William Godwin and Samuel Johnson: "Wollstonecraft's sentimental anti-Jacobin novels most resemble Godwin's sentimental anti-Jacobin novels... whereas her essays most resemble Johnson's writings" ("Gender and Cultural Analytics: Finding or Making Stereo-

types?" par. 29). Her analysis creates new categories that draw gender into genre—squaring the gender binary. She creates doubly refracted categories like, "men writing as men," "women writing as women," "women writing as men," "men writing as women" (par. 35).

15

When using reproducible tools, it bears asking what is being reproduced. Applying text analysis to racial categories, Edwin Roland and Richard Jean So explore whether a machine can correctly identify the race of a novel's author. The point, they are careful to say, is not to test whether the machine is correct, but to examine how the categories themselves are constructed. As they explain, "the machine simply measures the robustness of the social constructedness of these given categories and points out what gives them such vigor" ("Race and Distant Reading" 69). Analyzing a large corpus of novels by white and black authors, So and Roland find that black authors generally display more varied vocabulary than white authors. From this result, they infer that White authorship, as a category, only coheres *as a category* against the variance of Black authorship. In other words, White authorship is only visible as a category when it is compared to Black authorship, which displays overall more varied uses of language forms.

Interestingly, this quantitative exercise points Roland and So toward a peculiarity in the results: that the algorithm wrongly categorizes Black-authored novel, James Baldwin's *Giovanni's Room* (1956), as being written by a White author. This misclassification is due to a single word—"appalled"—whose usage, according to the model, has generally been a signal of White authorship. The term only occurs once in the text, in an early scene where the narrator David describes his strained relationship to his father: "I did not want to be his buddy. I wanted to be his son. What passed between us as masculine candor exhausted and appalled me" (Rpt. in Roland, So 71; emphasis mine). Given the connotation of whiteness in "appalled" and its middle French root, "apalir" (to grow pale), Roland and So explain that this term—which emerged as a result of the computer's mistake—enables their analysis to coordinate gender to race: "the moment David develops a troubled relationship to normative masculinity [as] also the moment he becomes 'White'" (71). Rather than reify racial categories, this computational experiment points out that they are constructed on flimsy terms.

16

So and Roland began with the binary form, the binary categories of Black/White, but they ended with its fracturing. This fracturing, evident in the computer's misclassification of authorship in *Giovanni's Room*, leads to a reading which coordinates race with gender and sexuality. This "queering

the machine's color line," the authors explain, opens up new readings of intersectionality in the novel which "in their current form, our data and model are not robust enough to handle" (72).

Rather than reproducible, this method is refractive, reflecting (and corrupting) the constructed nature of binary structures back at the critic. This method, as I explain in the next section, takes an *iterative* stance toward gender forms in literary data.

1.3 Iteration in gender

1

Multiple text analysis projects draw from Butler's concept of performativity. In particular, they take the language of performance and qualities of the performative to describe and conceptualize their methodologies. For example, Mandell's project, introduced above, takes the behavioral quality of gender performativity and associates it with authorial style, asserting that both gender and genre "are... highly imitable," so that "anyone can adopt gendered modes of behavior, just as anyone can write in genres stereotypically labeled M/F" (par. 30). Bode applies the "performative" to illustrate the analytical act itself, which necessarily intervenes and becomes inseparable from data as object of analysis, where "performativity" is "attuned to the coconstitution of computational methods and objects" ("What's the Matter with Computational Literary Studies"). Her project, *Reading at the Interface*, explores how "meaning is not carried by texts but produced in situated interactions between readers, texts and contexts," including computational tools.⁶ Another project, from the field of Computational Humanities, Kent K. Chang, Anna Ho and David Bamman cite Butler's performativity as motivation to computationally "measure subversion" in TV show dialogue. Their uses a language model to predict whether samples of a TV show, *The Big Bang Theory*, are queer-coded. If the model marks certain sections of dialogue between heterosexual characters as queer, they argue, then it will have indicated moments of "subversion" in the show (Chang et al.).

2

Unlike those projects, however, this project draws from performativity's methodology in depth. In doing so, it demystifies a common misreading of performativity—that it denotes an act or series of acts that can be imitated at will. Indeed Butler's follow up to *Gender Trouble: Feminism and the*

⁶Bode, Katherine. "Data beyond representation: From computational modelling to performative materiality." MLA Convention, Seattle, 10 January 2020. <https://katherinebode.wordpress.com/home/mla-convention-2020/>

Subversion of Identity (1990), aimed to clarify how performativity is bound by social systems and norms in ways that delimit agency. In this work, *Bodies that Matter: on the Discursive Limits of Sex* (1996), Butler explains that gender performativity is a compulsory process that constitutes the subject. In Butler's reasoning, the body only materializes (*matters*) through the repetition of gender norms. And it is through repetition that the body also assumes subjectivity. Crucially, "there is no subject who decides on its gender. . . . on the contrary, gender is part of what decides the subject" (ix). In other words, gender is an activity that determines both the body and subjectivity. It is from this context of a compulsory gender system where the subject must find a way to re-work the terms of expression.

3

Butler's argument for performativity draws from poststructuralist feminist theorist Luce Irigaray. Irigaray is best known for her readings of phallogocentrism (the idea that the phallus is a focus or lense for understanding other phenomena), in the works of classical and modern philosophers, like Plato, Aristotle, Descartes, and Kant. Irigaray finds that these philosophers represent the feminine through masculine terms, "on the basis of masculine parameters," as she describes in her book, *The Sex Which Is Not One* (Irigaray 23). Her conceptualization of the feminine's circumscription within the gender binary directly influences Butler's conceptualization of performativity and how it might resist rigid signficatory systems (18).

4

Irigaray's analysis of the feminine centers on its unrepresentable status within dominant framings like the gender binary. She argues that the binary of male/female effectively *precludes* the representation of the feminine: "The economy that claims to include the feminine as the subordinate term in a binary opposition of masculine/feminine excludes the feminine, produces the feminine as that which must be excluded for that economy to operate" (Butler 10). While the feminine is effectively excluded from the system, it conditions the representation of terms *within* the system. This excluded feminine, this "constitutive outside," makes possible the creation of the inner feminine, the "domesticated" feminine. It makes legible an internal feminine. She, what Butler calls the "unspeakable" feminine, is the *matter* from which the binary may come into intelligibility.

5

This external feminine is *matter*: the generative, creative material from which distinctions emerge. Butler explains that,

every explicit distinction takes place in an inscriptional space

that the distinction itself cannot accomodate. Matter as a *site* of inscription cannot be explicitly thematized. And this inscriptional site or space is, for Irigaray, a *materiality* that is not the same category of "matter" whose articulation it conditions and enables. . . this second inarticulate "matter" designates the constitutive outside of the Platonic economy; it is what must be excluded for that economy to posture as internally coherent" (emphasis original, 12-13).

Irigaray's association of this excessive feminine with "matter" draws simultaneously from a classical analogy between the form/matter and masculine/feminine, in the Aristotelian tradition, and from the etymological association between the feminine as *mater*, mother, and significantly, as *matrix*, the generative material for all life. It is this *matrix*, this "inscriptional space", which cannot be articulated, but from which all distinctions of male/female can emerge.

6

How is this feminine to enter into representability? How can it be invoked directly without subscribing itself to the ruling structure? Or, in Butler's formulation, "how can one read a text for what does not appear within its own terms?" (*Bodies* 11). As a constitutive exclusion which makes the intelligible binary possible, the excluded feminine cannot be represented in the same terms (10).

7

The answer, as it turns out, involves breaking open signficatory procedures. Butler considers the referential power of language: "Can language simply refer to materiality, or is language also the very condition under which materiality may be said to appear?" (6). For example: the physical body. In order to refer to a physical body, language must first assume a body. That is, the body must be posed *in language* first. Therefore, Butler reasons, "signification produces as an effect of its own procedure the very body that it nevertheless and simultaneously claims to discover as that which precedes its own action" (6). In their characteristically clausey prose, Butler explains that signification *creates* the body which it appears to reference. Which leads them to a major realization: "the mimetic or representational status of language. . . is not mimetic at all. On the contrary, it is productive, constitutive, one might even argue performative' (*Bodies* 6).

8

Performativity, then, denotes the perpetual creation and recreation of norms, a discursively-based, signficatory cycle which *produces* that which it

appears to reference. As they explain, "performativity must be understood not as a singular or deliberate 'act,' but, rather, as the reiterative and citational practice by which discourse produces the effects that it names" (xii). According to Butler, it is through this discursive performativity that "sexed" bodies emerge: "'Sex' is not a simple fact or static condition of a body, but a process whereby regulatory norms materialize 'sex' and achieve this materialization through a forcible reiteration of those norms" (xii). What appears as physical or biological category, "sex," is effectively delimited by and through a discursive framing. Bodily materiality is a "sedimentation" formed through repetition, whereby meaning is signified and re-signified endlessly, and never completely.

9

Categories like sex and gender, as Butler explains, are never stable, but require constant repetition:

This "being a man" and this "being a woman" are internally unstable affairs. They are always beset by ambivalence precisely because there is a cost in every identification, the loss of some other set of identifications, the forcible approximation of a norm one never chooses, a norm that chooses us. (86)

Repetition stabilizes gender identity against a field of potential exclusions, what Butler calls the "cost in every identification." Every identification creates this field of exclusion. And it is this material which will come back to destabilize coherence. With an ominous overtone, Butler explains that, "The political terms that are meant to establish a sure or coherent identity are troubled by this failure of discursive performativity to finally and full establish the identity to which it refers" (Butler 140).

10

This necessity to repeat gender signification is a weakness, which Butler will use as an exploit. For, within this signifiatory circle, lies the possibility of *resignifying* terms. Resignification happens by repeating what has been excluded from the signifiatory economy, "citing" what Butler calls a "repudiated" meaning. As Butler explains, every "identification takes place through a repudiation which produces a domain of abjection (xiii). This domain of abjection, containing identities that are repudiated, offers "a field of disruptive possibilities" for resignification (Butler 10). Butler explains, "this disavowed abjection will threaten to expose the self-grounding presumptions of the sexed subject" (xiii). Through this threat, "what has been foreclosed or banished from the proper domain of "sex"... might at once be produced as a troubling return" (Butler xxx).

If the troubling return is repeated continually, Butler argues, the stability of the original identity will be brought into question. Butler offers an example with the term "queer," which is a term that harnesses its own history of repudiation for political purposes, "resignifying the abjection of homosexuality into defiance and legitimacy (Butler xxviii).

The compulsion to repeat an injury is not necessarily the compulsion to repeat the injury in the same way or to stay fully within the traumatic orbit of that injury. The force of repetition in language may be the paradoxical condition by which a certain agency—not linked to a fiction of the ego as master of circumstance—is derived from the impossibility of choice. (183)

It is through the force of repetition that repudiated meanings, which necessarily draw from insulting and painful pasts, come to alchemize new significations. The paradox is that what appears to be a cyclical or even robotic behavior takes on an agentic capacity. Automata, at some point, attain agency.

Subversion thus reaches toward a destabilization of the ruling system. And this is where Butler's theory is often misunderstood. For example, Chang et. al's text analysis project applies the label of "subversive" to queer-coded characters within an otherwise heterosexual TV show, claiming that: "gender performativity provide[s] the essential framework for understanding how characters on TV subvert traditional gender roles and expectations" (4). Their analysis finds that two of its two of the main characters from *The Big Bang Theory*, Raj and Howard, share dialogue that is romantic and intimate in nature. The language model tags many instances of their dialogue as dialogue between partners. And indeed, these characters display an atypical intimacy which is the target of many of the show's jokes. The authors argue that "subversion," for this project, emerges in "an undercurrent of emotional intimacy and dependency that challenges the rigid boundaries of what is considered acceptable" (Chang et al., 10). In their conclusion, they posit that, "Raj and Howard offer more than occasional punchlines; they represent an instance of queer love: they have to invent a form of togetherness for themselves" (Chang et al., 2024, p. 9). However, according to Butler, subversion is more than exhibiting a non-dominant norm. It is doing so in a way that challenges established norms. Butler carefully stipulates against instances of apparent subversion which effectively prop up heterosexual regimes:

[T]here are forms of drag that heterosexual culture produces for itself—we might think of Julie Andrews in *Victor, Victoria* or Dustin Hoffmann in *Tootsie* or Jack Lemmon in *Some Like It Hot* where the anxiety over a possible homosexual consequence is both produced and deflected within the narrative trajectory of the films. These are films which produce and contain the homosexual excess of any given drag performance, the fear that an apparently heterosexual contact might be made before the discovery of a nonapparent homosexuality. This is drag as high het entertainment, and though these films are surely important to read as cultural texts in which homophobia and homosexual panic are negotiated, I would be reticent to call them subversive.
(85)

Despite being queer-coded, these characters remain within heterosexual relationships. As a result, their queerness effectively "produce[s] and contain[s] the homosexual excess" within the parameters of the show. They use humor to temper these excesses. Borrowing from Irigaray, they are an example of "domesticated" queerness, rather than subversive queerness.

13

Subversion always aims toward destabilization. In doing so, repetition is key, enabling the introduction of the repudiation back into the system. Returning to Irigaray's writing, Butler explains that Irigaray achieves subversion by "mim[ing] philosophy... and, in the mime, tak[ing] on a language that effectively cannot belong to her" (12). Butler cites a passage where Irigaray *mimes*, or repeats, the language of Plato. Her repetition, however, has a particular effect: she "performs a repetition and displacement of the phallic economy" (Butler 18). In an interesting double-level inhabitation, Butler approximates what Irigaray must be feeling to inhabit Plato's written style:

I will not be a poor copy in your system, but I will resemble you nevertheless by miming the textual passages through which you construct your system and showing that what cannot enter it is already inside it (as its necessary outside), and I will mime and repeat the gestures of your operation until this emergence of the outside within the system calls into question its systematic closure and its pretension to be self-grounding." (*Bodies* 18)

Irigaray undermines the authority of the masculine through repetition, by "cit[ing] Plato again and again, but the citations expose precisely what is

excluded from them, and seek to show and to reintroduce the excluded into the system itself" (Butler 18). Butler explains that "This is a taking of his [Plato's] place, not to assume it, but to show that it is *occupiable*... a kind of overreading which mimes and exposes the speculative excesses in Plato (emphasis original, 11).

14

Through repetition, Irigaray displaces the logic of phallogocentrism, introducing something external to the system while simultaneously remaining within its terminology. Deception emerges from resemblance, and insubordination through subservience. The key is repetition, or more precisely, what Butler refers to as *iteration*, a continual miming of the authorizing norm. Through the iterative process, one brings back again and again the repudiated meaning as a troubling return. "The task," according to Butler, "will be to consider this threat and disruption... as a critical resource in the struggle to rearticulate the very terms of symbolic legitimacy and intelligibility" (xiii).

1.4 Iteration in python

1

Iteration is a core concept in programming. Iteration means repeating the same action to each bit of data in a collection of data. It drive one of the most important programming structures, called the "loop," which *iterates* through collections of data like text, performing the same operation to each word. Iterating or "looping through" words in a text will count their features. And counting, as I explain below, is the foundation of all text analysis tasks, from word clouds to text generation.

4

Before counting text, a text must first be standardized or "cleaned." The looping mechanism quickly cleans a text and remove extraneous or undesirable elements analysis. Elements that are highly frequent, like articles, prepositions, and pronouns, can drown out other terms which are considered more substantial, like nouns and verbs. Additionally, irregular cases also need to be standardized, because capital letters would count toward different words (for example, "talk" and "Talk" would be considered two different words). For similar reasons, word endings and inflections like plural forms (endings in "s") and tenses ('walked', 'walks' or 'walking') are converted into a root form. Once the text has been regularized, then the counts will be more accurate to its content.

Using a loop, a person may *iterate* through each word in a text, check

whether it needs to be "cleaned," and then perform a set of cleaning actions. For example, a first round of cleaning might check for the presence of punctuation and capital letters. Below is an example of a `for` loop that filters out punctuation and transforming any capital letters into lowercase forms:

```
more time to run.  
clean = []  
for word in text:  
    if word.isalpha():  
        clean.append(word.lower())
```

3

The loop begins by creating an empty placeholder, `clean`, where words will be saved in their clean form. after passing the filters below. The next line begins the `for` loop, which iterates through each word in the text, in our case, in Woolf's novel, *Orlando*. The third line contains the first filter, `isalpha()`, which checks if the word is comprised of only alphabetic characters (i.e. it is not a number or punctuation). If the word is alphabetic, it passes the filter. The word moves to the next line, where it is lowercased, `lower()` and finally added to `clean`. When the loop is done running over every word in the text of *Orlando*, the final list of words in `clean` will only contain alphabetic and lowercased letters.

5

The next task in cleaning is to remove what are called `stops` or "stop words," which consist of prepositions, articles, pronouns, and auxiliary verbs.

```
["i", "me", "my", "myself", "we", "our", "ours", "ourselves", "you",  
 "your", "yours", "yourself", "yourselves", "he", "him", "his",  
 "himself", "she", "her", "hers", "herself", "it", "its", "itself",  
 "they", "them", "their", "theirs", "themselves", "what", "which",  
 "who", "whom", "this", "that", "these", "those", "am", "is", "are",  
 "was", "were", "be", "been", "being", "have", "has", "had", "having",  
 "do", "does", "did", "doing", "a", "an", "the", "and", "but", "if",  
 "or", "because", "as", "until", "while", "of", "at", "by", "for",  
 "with", "about", "against", "between", "into", "through", "during",  
 "before", "after", "above", "below", "to", "from", "up", "down", "in",  
 "out", "on", "off", "over", "under", "again", "further", "then",  
 "once", "here", "there", "when", "where", "why", "how", "all", "any",  
 "both", "each", "few", "more", "most", "other", "some", "such", "no",
```

```
"nor", "not", "only", "own", "same", "so", "than", "too", "very", "s",  
"t", "can", "will", "just", "don", "should", "now"]
```

Due to their high frequency and low semantic value in comparison to nouns, verbs, and adjectives, and adverbs, stop words will skew the results of analysis, so they must be removed.

```
no-stops = []  
for word in cleaned:  
    if word not in stops:  
        no-stops.append(word)
```

6

The loop takes each word the `clean` list, and checks to see if that word is also contained in above list of `stops`. If the word is not a stop word, then it will be added to a new list, `no-stops`. After this processing, the following words remain from the first sentence of Orlando:

```
['could', 'doubt', 'sex', 'though', 'fashion', 'time', 'something',  
'disguise']
```

7

As mentioned in this chapter's introduction, the gender pronouns, "He" and "his", as well as the em dash following "he" has been removed.

The final step of cleaning involves stripping word inflections to get the word's root form, or "lemma". This process involves looking up each word, one by one, to return the shortest lemma found. It uses the WordNet database, an electronic lexical database used for computational linguistics. WordNet contains every word's root form, as well as "synsets" or synonym sets, representing underlying relationships between words.⁷ In the python code, the lemmatizer function `lemmatize()` is called from the `WordNetLemmatizer` module.

```
lemmatized = []  
for word in no-stops:  
    lemma = WordNetLemmatizer.lemmatize(word)  
    lemmatized.append(lemma)
```

⁷Christiane Fellbaum (1998, ed.) WordNet: An Electronic Lexical Database. Cambridge, MA: MIT Press.

8

The above code iterates through the list of **no-stops** (which has already been stripped of punctuation, numbers, capitalization, and stop words) and lemmatizes each individual word using the `lemmatize()` function. It saves the word to a new variable, called **lemma**, which it finally appends to the list called **lemmatized**. After iterating through every word in the no-stops list, and lemmatizing each one by one, the text is ready for analysis.

8.1

Text analysis is, at a most basic level, about counting words. From counting how frequently a word appears, one can start to derive patterns about word usage.

In python, text analysis tasks are based on a module called **Counter**, which contains the functions for starting and keeping track of word counts. The logic, which is simplified below from the source code, (written originally in the C programming language⁸) keeps track of how many times each word has appeared while going through a text one word at a time.

```
counts = {}

for word in text:

    if word not in counts:
        counts[word] = 0

    counts[word] = counts[word] + 1
```

For every word, the program checks whether it has already been seen before, starting with the statement `if word not in counts`. If it has not been seen, the program runs the indented line below to set the count to zero. Then, moving to the following line, the count is increased by one to represent the current appearance. However, if the word *has* already been seen, the program will skip the 0 line and instead retrieve the existing count, to increase it by one again. By repeating this process for every word, the program builds a complete record of how many times each unique word occurs.

The counts for the first sentence of the text are the following:

⁸link to source Counter (python) code: https://github.com/python/cpython/blob/main/Lib/collections/__init__.py; link to source code in C language: https://github.com/python/cpython/blob/main/Modules/_collectionsmodule.c

```
Counter({'could': 232,
        'doubt': 27,
        'sex': 53,
        'though': 94,
        'fashion': 11,
        'time': 154,
        'something': 73,
        'disguise': 4,
        ...})
```

The logic of this counting loop is fully articulated in another programming language, the C language. The C language is a lower-level language that underlies Python code. In that language, the instructions for the same loop appear with the following syntax.

```
while ((key = PyIter_Next(it)) != NULL) {

    oldval = PyDict_GetItem(mapping, key);

    if (oldval == NULL) {
        newval = 1;
    }
    else {
        newval = oldval + 1;
    }

    PyDict_SetItem(mapping, key, newval);
}
```

The C loop, which begins with `while`, updates a value by one: `newval = oldval + 1`. While the code in the C language is more convoluted than Python, it runs more efficiently. This is because C code can run at lower levels of the software stack, which is where computation and counting actually occurs. Python, by contrast, needs to be translated ("compiled") in order to be understood by lower-level machine processes.

The text analysis library Natural Language Toolkit, or NLTK, is a python library, or collection of code, written specifically for working with language data. Due to its modular structure, it's a good library for beginners in Python (it was the first library that I ever learned). NLTK comes packaged

with useful functions for simple text analysis tasks, like `q concordance()`, which displays the context window surrounding a given word, and `similar()`, which finds words that are used in similar contexts to a given word.

For example, running `concordance()` on the word "woman" will output sentences that contain that word from *Orlando*:

Displaying 15 of 187 matches:

```
ty manly charm quality old woman loved failed growing old w
en without knowing perhaps woman heart intricate ignorance
indow pulled among cushion woman laid worn old made bury fa
red dyed cheek scarlet old woman loved queen knew man saw o
finger swore vilely master woman scarcely le bold speech le
ment panic twelve poor old woman parish today drink tea ton
perverse cruel disposition woman broke engagement night eve
verladen apple old bumboat woman carrying fruit market surr
embassy figure whether boy woman loose tunic trouser russia
ed stared boy ala boy must woman could skate speed vigour s
me standstill handsbreadth woman orlando stared trembled tu
d asked tumult emotion old woman answered skin bone trulls
and longer full prying old woman said stared one face bump
save sea bird old country woman hacking ice vain attempt d
ce melt heat pity poor old woman natural mean thawing must
```

10

This context window reveals a text that has been stripped of punctuation, capital letters, prepositions, articles, pronouns, auxiliary words. What remains is largely nouns, verbs, and adverbs. The task is to then count and analyze these remains.

11

For that purpose, another function, called `similar()` displays words that are used in similar contexts of the target word. This function takes a canonical concept from NLP called "distributional similarity," where words that appear in similar contexts tend to have similar meanings. For woman, those words are the following,

```
['man', 'moment', 'night', 'boy', 'word', 'world', 'child' 'pen',
'ship', 'door', 'one', 'room', 'window', 'light', 'little', 'lady',
'table', 'book', 'queen', 'king']
```

To compute the results of `similar()`, the program first takes the context of the target word from `concordance()`. Then, it counts each of the words

surrounding woman. As seen above with the `Counter()`, it keeps a running tally of these words. Then, it searches other words in the database and, again, takes a count of surrounding words for these words. It then looks for words that have the most shared contexts, that is, the most shared words within their context window, with the word woman. Finally, it outputs a list of terms that share the most contexts with the word.

12

For the program then, word meaning is computed through patterns from its usage based on word counts. This concept is a foundational one in all text analysis going into deep learning. Importantly, the computer does not understand or impute *meaning* to the words. Rather, it only counts each word as a string, that is, as a piece of data composed of alphanumeric sequences. From there, it derives meaning from patterns of usage.

13

That is, in a nutshell, what so-called *deep learning* methods do. Deep learning methods, like those that drive generative text processes (discussed in detail in chapter 5) move from displaying the results of word counts into generating word predictions.

Many of these methods are based on the concept of "word embeddings" or "word vectors" to ascribe machine-interpretable meaning to strings. Like `similar()` and `concordance()`, word embeddings work from patterns of word similarity based on context. Unlike the previous methods, however, word embeddings encode numeric values to a given word based on its context. They are called "vectors" because these numeric values represent points in a graph. By graphing word similarity, word vectors translate relationships of word context into geometric space.

14

A vector for a single word, "woman," like the one below, will contain a list of numbers that represent a high similarity score between "woman" and another word.

```
['child', 0.9371739625930786,  
'mother', 0.9214696884155273,  
'whose', 0.9174973368644714,  
'called', 0.9146499633789062,  
'person', 0.9135538339614868,  
'wife', 0.9088311195373535,  
'being', 0.9037441611289978,  
'father', 0.9028053283691406,  
'guy', 0.9026350975036621,
```

```
'known', 0.8997253179550171,  
...]
```

Based off a popular model trained from twitter posts, the vector which represents “woman” contains a list of numbers that score the similarity of the word “woman” to other words in the dataset. Here, the word “woman” is most closely associated to the word “child,” with a similarity score, or “weight,” of .93, or 93%, then with “mother,” with .92, then “father,” with .90, and so on.

15

Commonly, word vectors are organized into a matrix, or tabular, format. Using this matrix format, further mathematical operations are possible using statistics, linear algebra, and calculus—the building blocks of deep learning methods.

The inventors of word vectors introduced their power to the world via a formula that uses math to compute language meaning (Mikolov et al. 2013). The famous formula uses a mathematical equation to calculate the meaning of the word "Queen". Here, the vector for the word "queen" is calculated by subtracting the vectors of "man" from "king," and finally, adding the vector for "woman."

$$\text{vector("Queen")} = \text{vector("King")} - \text{vector("Man")} + \text{vector("Woman")}$$

The paper, whose insights about technology of word vectors made Large Language Model development possible, began with this formula. It’s cross-cultural intelligibility, therefore, cannot be understated. Gender, which is necessarily imbricated with imperialist concepts, becomes an operand for computing language meaning. Like Ted Underwood’s graph which puts gender into an oppositional relationship, gender is the *model* which shows the algorithm to be true.

The word vector for "Queen consists of numbers, a long list of numbers.

```
'queen': -0.3583, -0.6767, 0.1689, -0.6164, -0.0135, -0.2284, 0.3359,  
0.1149, -0.1512, -0.8014
```

Within the long list of probabilities, each individual probability score represents that word’s contextual relationship to another word in the dataset. One of these numbers may represent Queen’s relationship to the word "Woman," and this number will have a higher probability score than that which represents the relation to the word "man." These long chains of numbers, which encode these probability scores between "Queen" every other individual word

in the database, function effectively like the word's definition. In kind of literalization of the poststructuralist floating signifier, a single word is defined as, not what it means in itself, but how it relates to (and differs from) every single other word. In word vectors, then, meaning emerges from association, from proximity and distance, between words.

Like Butler's matter or *matrix* which produces the binary of masculine/feminine, here is another matrix that produces meaning according to masculinist parameters. The matrix format of the word vector enables a host of mathematical operations and algorithms: from example, cosine similarity calculates the distance between two vectors, that is, two words; and regression analysis guesses the relationship between an independent and dependent variable, or relation between one word and another.

16

In the following section, I use these word vectors as a starting point to explore terms related to gender terms in *Orlando*, starting with the terms "woman" and "man." Although I start with the gender binary, word vectors enable me to quickly proliferate beyond it. Rather than resolving gender into an oppositional structure or a neat formula, it cracks open.

1.5 TODO Fluidity

1 Woolf's novel, *Orlando: A Biography*, is ideal for a computational study of gender for two reasons. First, published in 1928, it is an early example of transgender narrative. Second, as many critics have noted, its characteristic modernist experimentation with the limits of language destabilizes normative concepts of identity and gender. Jane de Gay, Jill Channing, and Christy L. Burns, for example, assert that Woolf deploys imaginative elements, magical realism, and parody, respectively, to resist realist narrative expectations. De Gay describes Woolf's writing as "feminist historiography" that "reject[s] Victorian patriarchal metanarratives" and instead "use[s] the strategies of fiction to bring history alive and make it live in the present" (de Gay 71). In a similar vein, Burns and Channing both point out that Woolf uses fantastical elements to disrupt expectations of plot and narrative to challenge the stability of gender and identity. Doubling down on the role of language, some critics emphasize that Woolf's narration purposefully obfuscates any coherence between gender, identity, and even race and nationality. For example, Victoria L. Smith asserts that "the fantastic content in the novel is directly linked to the undecidability/impossibility of the form of the novel and of the protagonist" (58). Pamela Caughie agrees, arguing that *Orlando*'s transgressiveness comes from its discursive moves: "Far from

defeating sexual difference, as many feminist critics claim, Orlando enacts it, enshrines it, exploits it, makes a spectacle of it, but as a playful oscillation not a stable opposition” (Caughie 48).

2 In what follows, I pursue a text analysis method that I call “iterative” for the way it moves between close and distant reading, a process similar to what Andrew Piper calls “bifocal” reading. This process, in Piper’s words, “no longer us[es] our own judgments as benchmarks. . . but explicitly construct[s] the context through which something is seen as significant (and the means through which significance is assessed)” (17). My method takes the output of computations as material for close reading analysis, using the computer to identify words that I then examine in context and with critical subjectivity.

3 I begin with a list of terms that are computed as similar to “woman” and “man” in the text. To get distinctive results for each gender, I modified the code to remove any words with strong associations to the opposite gender. For example, I compute words that are most positively associated with “woman” and most negatively associated with “man.” Though this analysis, like Underwood’s, begins with a binary formulation of gender, this binary quickly destabilizes as I move deeper into the close reading analysis. By iterating through distant and close reading, the terms swell with significations that pluralize the binary and work toward resignifying the initial results for “woman” and “man.” Below are the words most closely associated with “woman” and “man,” respectively, each with its own score, or probability. The following are the words most associated with “woman”:

```
[('soft', 0.3692586421966553),
('named', 0.34212377667427063),
('sciatica', 0.3223450779914856),
('frilled', 0.3187992572784424),
('despaired', 0.31375786662101746),
('friend', 0.31238242983818054),
('delicious', 0.30853813886642456),
('winked', 0.30514153838157654),
('notion', 0.3047487139701843),
('seductiveness', 0.30290719866752625)]
```

And the following with “man”:

```
[('chequered', 0.4025157392024994),
('fact', 0.3394489586353302),
('denounced', 0.3346075117588043),
```

```
(‘house’, 0.33423593640327454),
(‘curiosity’, 0.33144116401672363),
(‘defend’, 0.3284823000431061),
(‘dancing’, 0.3282632827758789),
(‘marbling’, 0.3184848427772522),
(‘cynosure’, 0.3057470917701721),
(‘rather’, 0.3024100363254547)]
```

4 At first glance, the terms for each list appear to align with existing conceptions of femininity and masculinity, such as “soft” and “frilled” for “woman,” and “fact” and “defend” for “man.” But as I examine these words in context, the gender binary becomes less clear cut. From the “woman” list, the term “delicious” only appears after Orlando has transitioned into a woman in the story. As I explain below, this term reveals a relationship to queerness characterized by a distinctly passive mode of gender subversion. Then, from the “man” list, I examine a term that only appears once in the novel, at a significant moment which initiates what I call the novel’s “crisis of signification” with language. Following this thread, I examine further passages that develop this theme as it spreads into Orlando’s interior narration. I conclude with a close reading of a rather dramatic passage that contains words from both the “woman” and “man” lists, and that coordinates this crisis with themes of gender ambiguity and homosexual desire.

5 I begin with the word “delicious,” which occurs only after Orlando transitions into a woman. Three of this term’s five occurrences appear in a single passage, when Orlando is sailing from Turkey back to her native England. The ship captain offers Orlando a slice of beef, which sends her into a rapturous speculation about the joys of womanhood:

‘A little of the fat, Ma’m?’ he asked. ‘Let me cut you just the tiniest little slice the size of your fingernail.’ At those words a delicious tremor ran through her frame. Birds sang; the torrents rushed. It recalled the feeling of indescribable pleasure with which she had first seen Sasha, hundreds of years ago. Then she had pursued, now she fled. Which is the greater ecstasy? The man’s or the woman’s? And are they not perhaps the same? No, she thought, this is the most delicious (thanking the Captain but refusing), to refuse, and see him frown. Well, she would, if he wished it, have the very thinnest, smallest shiver in the world. This was the most delicious of all, to yield and see him smile. ‘For nothing,’ she thought, regaining her couch on deck, and contin-

uing the argument, 'is more heavenly than to resist and to yield;
to yield and to resist. (114)

6 Here, "delicious" describes a refusal, then a yielding—the vacillations of what appears to be a passive form of pleasure, ostensibly opposed to the active pleasure of pursuit which Orlando enjoyed as a man. Although the word "delicious" describes an arguably feminine experience of pleasure, about withholding and, eventually, submitting to the active force, it is a pleasure rooted in what is not quite passivity and not quite power.

7 To get a deeper understanding of this term, I run another similarity search with "delicious" as the target word. The top result, the word most related to "delicious" in the text, is "culpable." I then turn back to the text to examine where this word appears, which happens twice, both times in the same scene on the ship, during Orlando's ruminations about the pleasures between the sexes. The first of those appearances occurs within one of the few sentences in the novel in which Orlando considers her homosexuality explicitly:

And as all Orlando's loves had been women, now, through the culpable laggardry of the human frame to adapt itself to convention, though she herself was a woman, it was still a woman she loved; and if the consciousness of being of the same sex had any effect at all, it was to quicken and deepen those feelings which she had had as a man. (119)

8 "Culpable," from the Latin "culpa," meaning fault, denotes a body deserving of blame for its refusal to conform. Here, the word modifies "laggardry," which describes the obstinacy or persistency of Orlando's love for women, despite that she is now a woman herself. It seems that, for Orlando, desire is defined by a sense of guilt about refusing to conform. The next usage of this term occurs soon after, when Orlando reprises her earlier thoughts about the deliciousness of refusal:

'To refuse and to yield,' she murmured, 'how delightful; to pursue and conquer, how august; to perceive and to reason, how sublime.' Not one of these words so coupled together seemed to her wrong; nevertheless, as the chalky cliffs loomed nearer, she felt culpable; dishonoured; unchaste, which, for one who had never given the matter a thought, was strange. (120)

9 Rehearsing the conventional roles of the sexes, roles which Orlando fails to fit into, she feels (in addition to "culpable") "dishonour[ed]" and

“unchaste”—words associated with an ideal of virtuous femininity. This feminizing language deepens the relationship between desire, guilt, and deliciousness, the last of which was previously characterized as pleasure in a passivity that is not quite powerless. Here, Orlando’s refusal to conform to social expectations, for which she feels culpable, constitutes another kind of passivity, a form of refusal. The connection between “delicious” and “culpable” seems to define queer desire as a distinctly passive mode of resistance.

10 Now, I move to Orlando’s experience as a man, returning to the original list of similar terms. I return to the list of vectors associated with the word “man,” and begin with “chequered,” which appears only once, at the start of the story, when Orlando makes his entrance, stepping into “the yellow pools chequered by the floor” (12). This moment is the first of many in which the narrator calls into question his credibility as a biographer, a self-described “scribe,” who distinguishes his role from the poet who embellishes and exaggerates reality through figurative language. However, the narrator’s commitment to straightforward description soon gallops into full-fledged figuration when he tries to describe Orlando’s beauty:

Directly we glance at Orlando standing by the window, we must admit that he had eyes like drenched violets, so large that the water seemed to have brimmed in them and widened them; and a brow like the swelling of a marble dome pressed between the two blank medallions which were his temples. Directly we glance at eyes and forehead, thus do we rhapsodize. Directly we glance at eyes and forehead, we have to admit a thousand disagreeables which it is the aim of every good biographer to ignore. (12-13)

11 Here, the narrator’s evocative language undermines his pretense to objectivity. This slip into figurative language eventually grows into a crisis of signification that recurs persistently through the novel. This crisis will eventually affect not only language’s ability to signify, but also that of gender. As Victoria L. Smith affirms, “what happens in the novel. . . and what it thematizes—language’s inability to adequately represent the ‘thing itself’—mirrors the undecidability of the text—is it a biography, an autobiography, fantasy, etc.—and the impossibility of the form of ‘woman’” (58).

12 The crisis of signification even spreads to Orlando’s internal thoughts, where it first emerges during a period of depression following his love affair with Sasha, a Russian princess. Here, I take the another term, “despaired,” from the list of vectors associated with “man.” Like “chequered,” this term occurs only once in the novel:

So then he tried saying the grass is green and the sky is blue and so to propitiate the austere spirit of poetry whom still, though at a great distance, he could not help reverencing. 'The sky is blue,' he said, 'the grass is green.' Looking up, he saw that, on the contrary, the sky is like the veils which a thousand Madonnas have let fall from their hair; and the grass fleets and darkens like a flight of girls fleeing the embraces of hairy satyrs from enchanted woods. 'Upon my word,' he said (for he had fallen into the bad habit of speaking aloud), 'I don't see that one's more true than another. Both are utterly false.' And he despaired of being able to solve the problem of what poetry is and what truth is and fell into a deep dejection. (75)

13 Orlando, deep in depression, struggles to understand the role of figuration in language. Much like the narrator who is suspicious of figurative language, Orlando also questions the truthfulness of figuration. First, he attempts plain language, "the sky is blue," "the grass is green," but these prove insufficient for describing a sky that moves "like the veils which a thousand Madonnas have let fall from their hair" and grass that "fleets and darkens like a flight of girls fleeing the embraces of hairy satyrs from enchanted woods." Orlando, who has just been abandoned by a woman, sees flight and modesty, qualities which he finds "false," in nature. The word "despaired" here, like "chequered" above, appears in a moment that elevates the role of figuration in language. It also marks a crossing of the crisis of signification over the diegetic threshold, into Orlando's internal world. These terms, which occur in scenes where the lines between fiction and reality, and internal and external narration are blurred, elevate the importance of language's imaginative capacities in the novel.

13.1 Man is divided until the female gender settles the division of the male.

Then, in chapter six, Orlando watches a toy boat negotiate a wavelet on the Serpentine river in London. The boat is momentarily submerged and then emerges on the other side of the wavelet. She associates this with the word "ecstasy", and hurries to telegram the phrase, 'a toy boat on the serpentine' and 'ecstasy', to her lover, who she knows will immediately understand what it means. As she hurries to the post office, she meditates on the nature of language and literature, which she now realizes is violently ecstatic. The toy boat now has the power to signify.

The symmetry of this episode with the last in the grass shows how Orlando's new perspective on language, her faith in language, endows it with

the power to mean.

14 In a final example, language's ability to signify becomes closely coordinated to gender. I examine the co-occurrence of words from both the "woman" and "man" lists within a single passage, the scene where Orlando first meets his beloved Sasha. The words "curiosity," which is associated with "man," and "seductiveness," which is associated with "woman," appear in a tumultuous meditation on Sasha's gender incomprehensibility. The drama begins when Orlando, upon seeing Sasha for the first time, cannot tell whether she is a man or a woman:

He beheld, coming from the pavilion of the Muscovite Embassy, a figure, which, whether boy's or woman's, for the loose tunic and trousers of the Russian fashion served to disguise the sex, filled him with the highest curiosity. The person, whatever the name or sex, was about middle height, very slenderly fashioned, and dressed entirely in oyster-coloured velvet, trimmed with some unfamiliar greenish-coloured fur. But these details were obscured by the extraordinary seductiveness which issued from the whole person. Images, metaphors of the most extreme and extravagant twined and twisted in his mind. He called her a melon, a pineapple, an olive tree, an emerald, and a fox in the snow all in the space of three seconds; he did not know whether he had heard her, tasted her, seen her, or all three together. (For though we must pause not a moment in the narrative we may here hastily note that all his images at this time were simple in the extreme to match his senses and were mostly taken from things he had liked the taste of as a boy. But if his senses were simple they were at the same time extremely strong. To pause therefore and seek the reasons of things is out of the question)... A melon, an emerald, a fox in the snow—so he raved, so he stared. When the boy, for alas, a boy it must be—no woman could skate with such speed and vigour—swept almost on tiptoe past him, Orlando was ready to tear his hair with vexation that the person was of his own sex, and thus all embraces were out of the question. (27-28)

15 The passage constellates the crisis of signification within the larger issue gender ambiguity. Orlando uses seemingly arbitrary metaphors, "A melon, an emerald, a fox in the snow," indicating that at the same time which he cannot place Sasha's gender, he also cannot find the right words to describe her. As Sasha's gender oscillates between male and female throughout

the passage, so the narrative voice alternates between Orlando's interiority and the narrator's commentary. Taking this crisis of signification beyond the diegetic narrative, the narrator's "pause" signals the constructed nature of the scene. This constructed quality is emphasized by the narrator's explanation for Orlando's choice of words, which have no "reason," and "were mostly taken from things he had liked the taste of as a boy." Language, like gender, thus becomes a tool for representation and performance.

16 The iterative reading practice, which bases close attention to language from the results of distant reading analysis, brings to the surface the relationship between gender and imagination, particularly imaginative forms of language, in the novel. As Pamela Caughie asserts, Orlando's gender transgression is intimately connected to the narrator's experiments in figuration and form: "Woolf brings out the arbitrariness of [sexual] identity, the arbitrariness of language itself, through Orlando's switching from one sex to the other, and from one poetic language to another, as well as through the shifting of her own rhetoric in this novel (42).

17 The list of words offer a starting point for isolating passages throughout the text, which then offer opportunities for finding new words to read in context. As the reader moves from distant to close, from a word vector to its context, the meanings of certain words gain new significations. And what happens with language in the novel also happens with gender. The above passage, with its "switching" and "shifting" discourse, which asserts that word choices are arbitrary and flowing, implies that gender is also a fluid phenomenon.

1.6 Conclusion

0 The rest of the first sentence, "—was in the act of slicing at the head of a Moor which swung from the rafters" (11)."

"Beginning with an image of a decapitated Moor, Orlando appropriates the exoticized subjects of colonial discourse to create for its protagonist a sexuality that is not bourgeois or heterosexual. However, Orlando's playful but insistent production of white femininity allows the protagonist to pass as respectable and heterosexual by displacing her transgressive sexuality onto racial others, thereby masking Orlando's masculine-identified literary and sexual desires." (Hovey 398).

2 As a discursive phenomenon, performativity is situated within second-wave feminism and its deconstruction of the gender binary. Common critiques of Butler point out the limits of this theory for posing gender and sexuality as discursive. From the field of Trans Studies, Jay Prosser prob-

lematizes Butler's "deliteralization of sex," which he applies to Queer Studies more generally, for the way it elides the real-world concerns of the body's materiality and its precarious existence in cross-gendered identifications. This thread of discursivity, and its implications within a larger trajectory of Queer Studies, is picked up again in this chapter's conclusion.

1 Butler on proliferation. We do not want it.

3 The argument that Orlando's subversiveness is a discursive one, opens the text to numerous critiques, particularly from Trans Studies. According Jay Prosser, Woolf's experimentation with language and narrative form belies the embodied reality of transsexuality. He argues that "Orlando is not about the sexed body at all but the cultural vicissitudes of gender. As h/er narrative propels h/er through four centuries of history, Orlando is free to move beyond h/er body—quite queerly, to break through the limits of the flesh" (Prosser 168). By "the sexed body," Prosser means the physical body that exists in a physical world. Orlando's transgressiveness, Prosser argues, results from a play of language and literary form that elides the specificity and the lived reality of the "sexed body." Due to its "ambivalence, a wavering around transition," "a transformation of transition into new identity," its "easy androgyny," this text is transgender rather than transsexual (Prosser 169).

4 A decade later, Omise'eke Natasha Tinsley writes about the problem of gender fluidity as a metaphor. In her essay, "Black Atlantic, Queer Atlantic: Queer Imaginings of the Middle Passage," Tinsley argues for the imbrication of sexuality and race through the lens of the Black Atlantic. By sexuality, Tinsley does not necessarily mean "same-sex" desire, but relationships from the Middle Passage, that "mak[e] disruption to the violence of the normative order. . . connecting in ways that commodified flesh was never supposed to" (199). Reading for relation rather than desire, her critique re-works the trope of fluidity which, drawing from the ocean, "is not an easy metaphor or queer and racially hybrid identities but for concrete, painful, and liberatory experience" (192-193). For Tinsley, fluidity is an opportunity for "a return to the materiality of water to make its metaphors mean more complexly, shaking off settling into frozen figures" (212). Reading from Dionne Brand's book, *Map to the Door of No Return* (2001), on the Middle Passage, Tinsley theorizes fluidity as a "social liquidation," being stripped by the water, particulars of identity washed away in the current. She explains that "brown bodies are gender fluid not because they choose parodic proliferations but because they have been 'washed of all this lading, bag and baggage'" (209).

5 Tinsley's critique surfaces the ways that gender fluidity, as a trope for queerness, obscures the connotations of water, especially the physical

one of corrosion. While the topic of physical embodiment and racialization is one that I take up in my fourth chapter, in the concept of flesh, here I have tried to show how attention to iterativity in computer code, and the fluidity it inspires for close and distant reading analysis, surfaces the material components that undergird what is often presented as an immaterial or abstract processes within quantitative analysis.

6 Wraps up, points to future chapters.